AP Computer Science A
Ticket Out the Door
Set 7: Boolean Operators

Name _____ Period _____

Skill 7.1 Exercise 1	
Evaluate whether each of the following is true or false for the conditions below, x = 11 and y = 5	
((x < 10) AND (y = 6)	
((x < 10) AND (y = 5)	
((x > 10) AND (y ≠-3)	
((x < 10) OR (y = 5))	
((x > 10) OR (y = 5))	

Skill 7.2 Exercise 1	
Write each of the following expressions using proper java syntax	
((x < 10) AND (y = 6)	
((x < 10) AND (y = 5)	
((x > 10) AND (y ≠-3)	
((x < 10) OR (y = 5))	
((x > 10) OR (y = 5))	

Skill 7.3 Exercise 1	
Indicate what is printed	
System.out.println(false && false);	
System.out.println(true && false);	
System.out.println(false || true);	

Skill 7.4 Exercise 1	

Refer to the following code to evaluate what is printed.
 	
int x = 79, y = 46, z = -3;
double d = 13.89, jj = 40.0;
boolean b = true, c = false;	

System.out.println(true && !false);	
System.out.println(!b || c);	

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System.out.println((jj = = 40) && !false);	
System.out.println(x != 3);	
System.out.println(!(x != 3));	
System.out.println(c || !(d > 0));	
System.out.println(!!true);	

Skill 7.5 Exercise 1
Declare and initialize an integer a. Declare and initialize a different integer b. Declare a boolean c. Initialize c to !(a = = b). What is the value of c? True or false?

Skill 7.6 Exercise 1	

Refer to the following code to evaluate what is printed.
 	
int x = 79, y = 46, z = -3;
double d = 13.89, jj = 40.0;
boolean b = true, c = false;	

System.out.println(b && c || !c);	
System.out.println(x == y && !(z < 0) || b && c);	
System.out.println(x != y && y==z && b || !c);	
System.out.println(x > y || c || b && jj%4 != 0);	

Skill 7.7 Exercise 1	
Simplify the following	
!(A > B || B != A)	
! (A == B || (B >= C || B < A))	

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Skill 7.7 Exercise 2

Which of the following Boolean expressions are equivalent to the expression `num ≥ 15`?

Select two answers.

A `(num > 15) AND (num = 15)`

B `(num > 15) OR (num = 15)`

C `NOT (num < 15)`

D `NOT (num < 16)`

A `NAND` gate is a type of logic gate that produces an output of `false` only when both of its two inputs are `true`. Otherwise, the gate produces an output of `true`. Which of the following Boolean expressions correctly models a `NAND` gate with inputs `P` and `Q`?

A `(NOT P) AND (NOT Q)`

B `(NOT P) AND Q`

C `NOT (P AND Q)`

D `NOT (P OR Q)`

Skill 7.7 Exercise 3

The table below shows the value of an expression base on the values of input1 and input2

Value of `input1`	Value of `input2`	Value of `expression`
`true`	`true`	`false`
`true`	`false`	`true`
`false`	`true`	`true`
`false`	`false`	`true`

Write an expression in terms of input1 and input2 that would produce the output.

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